

Yangruirui ZHOU

Email: zhouon007@gmail.com

Tel: (+1) 857-763-9402

Webpage: zyrron.github.io

EDUCATION

Boston University, Boston, MA

09/2020-01/2026

Doctor of Philosophy (Ph.D.) in Computer Engineering

GPA: 3.91/4.00

Advisor: Douglas Densmore

Thesis: Constraint-Driven Design Automation Frameworks for Microfluidics and Scalable Biological Networks

Research Focus: AI-driven Design Automation, Graph Algorithms, and Software Infrastructure for Scientific Systems.

University of Electronic Science and Technology of China (UESTC), Chengdu, China

09/2016-06/2020

Bachelor of Engineering in Software Engineering (*Elite program*)

GPA: 3.94/4.00 (Rank: 6/740)

Thesis: Analysis of Postoperative Applications of Pose Tracking Algorithms.

RESEARCH SUMMARY

Ph.D. in Computer Engineering with expertise in machine learning, graph algorithms, optimization, and Machine Learning systems. Experienced in developing scalable software infrastructures, intelligent design automation systems, and ML-powered applications. Strong background in Python, C++, algorithm design, and deploying ML techniques to solve large-scale optimization problems.

AWARDS & HONORS

iGEM "Gold medal" (Student Mentor, wiki: <https://2024.igem.wiki/bostonu/>) 11/2024

China Youth Science and Technology Innovation Award (Nominated by UESTC) 05/2020

Most Outstanding Students of UESTC 2019 (成电杰出学生, **10/5000** in UESTC, **1/740** in department) 12/2019

iGEM "Gold medal" and "Best Software Project" (wiki: <https://2019.igem.org/Team:UESTC-Software>) 11/2019

Outstanding Graduates Award of UESTC 10/2019

"Wu Liang Ye" Enterprise Scholarship (**2/740** in Software Engineering department) 09/2019

TECHNICAL SKILLS

Core Expertise: Algorithm Design, Machine Learning, Graph Algorithms, Search, Graph-based Optimization, Dynamic Programming, Constraint Formalization, Scientific Software Engineering, Formal Verification, Compiler & DSL Design, AI-assisted Scientific Computing, Prompt Engineering

Machine Learning: PyTorch, TensorFlow, NumPy, Scikit-learn

Programming: Python, C/C++, Java, TypeScript, HTML, Linux Shell, LaTeX

Biological & Microfluidic Skills: CNC Milling, Laser Cutting, Fusion 360, Microfluidic Devices Manual Design

RESEARCH EXPERIENCE

Research Assistant | Boston University, CIDAR lab

Boston, MA | 09/2020-Present

Supervisor: Prof. Douglas Densmore

- **Neptune — AI-powered Microfluidic Design Automation System (CUHK collaboration)**
 - Developed a scalable software infrastructure for automated design workflows.
 - Designed compiler pipelines and graph-based intermediate representations.
 - Implemented optimization frameworks supporting large-scale workflow automation.
 - Built modular software architecture supporting future extensibility.

- **Oriole — Graph Optimization Framework for Large-scale Biological Computation (*MIT collaboration*)**
 - Designed scalable graph optimization framework.
 - Implemented efficient graph partitioning algorithms.
 - Built reusable optimization library.
- **Vespa — Large-scale Automated Verification for Microfluidic Systems**
 - Developed an automated software verification framework.
 - Built scalable benchmark datasets with standard evaluation metrics.
 - Designed graph-based simulation pipelines.
- **Other Works**
 - Designed and fabricated modular microfluidic systems supporting multi-modal experimental workflows for studying cell–cell communication in controlled environments.
 - Integrated GPT-4, Claude, Gemini, DeepSeek, and Qwen into an intelligent microfluidic design workflow. Developed prompt engineering strategies for automated code generation. Built natural language interfaces enabling automated reasoning.

iGEM Team Student Mentor | BostonU (Wiki: <https://2024.igem.wiki/bostonu/>) Boston, MA | 05-10/2024
 Led computational and software design for iGEM 2024 project, bridging biological objectives with system-level modeling and implementation.

- Mentored students in system design and problem decomposition for biological engineering workflows.

iGEM Team Member | UESTC (Wiki: <https://2019.igem.org/Team:UESTC-Software>) Chengdu, China | 05-10/2019
 Built a cross-database sequence-to-protein mapping pipeline improving annotation between iGEM Registry and UniProt.

- Developed a ranking and filtering model improving alignment accuracy to >95% and increasing retrieval coverage by ~400%.

Student Research Internship | UCSB, Four “I”s lab Santa Barbara, CA | 06-08/2019
 Supervisor: Prof. Matthew Turk and Prof. Tobies Hollerer

- Reimplemented a state-of-the-art pose tracking algorithm for local evaluation and deployment.
- Benchmarked performance against a novel method using standardized datasets.

PROFESSIONAL SERVICES & ACTIVITIES

Mentor, EBRC Mentorship for Undergraduate and Master’s Students Program (EMUMS)	04/2025-Now
Direct Mentor, Eric Xie , Undergraduate Research Assistant, CIDAR Lab	06/2023-Now
International Workshop on Bio-Design Automation (IWBDA’25), Oral Presentation	09/2025
International Genetically Engineered Machine competition (iGEM’19, iGEM’24)	11/2019, 11/2024
Inventor/Contributor, Patent: Agricultural Heavy Metal Biosensor-Integrated Device	05-11/2024
Student Mentor of iGEM team: BostonU	04-10/2024
Chicago BioEngineering Conference (CBEC’24)	09/2024
Teaching Assistant, EC504: Advanced Data Structures and Algorithms , Boston University	01-05/2022, 01-05/2023
Engineering Biology Research Consortium (EBRC’23)	06/2023
International Workshop on Bio-Design Automation (IWBDA’22), Poster Presentation	10/2022

PUBLICATIONS

Google Scholar (<https://scholar.google.com/citations?user=Zel-iSQAAAAJ&hl=en>)

Journals:

[J1, IEEE-TCAD 44.2 (2024): 483-496.] **Zhou, Y.**, Oliveira, S. M., Sanka, R., McIntyre, D., & Densmore, D. (2024). Vespa: Logic-Level Constraint-Based Validation for Continuous-Flow Microfluidic Devices.

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[J2, Nature Chemical Biology 21.2 (2025): 268-279.] Padmakumar, J. P., Sun, J. J., Cho, W., **Zhou, Y.**, Krenz, C., Han, W. Z., ... & Voigt, C. A. (2024). Partitioning of a 2-bit hash function across 66 communicating cells.

[J3, IEEE-TCBB 22.4 (2025): 1888-1901.] **Zhou, Y.**, Voigt, C. A., & Densmore, D. (2025). Constraint-Based Sub-Graph Partitioning for Multi-Cellular Biological Networks.

Conferences & Workshops:

[C1, IWBD A'22] A Conceptual Interactive Microfluidic Design and Control Workflow

[C2, IWBD A'25] LaNVis: Biological Constraint-based Large Network Visualizer (Accepted for oral presentation)

Manuscripts in Preparation:

[J4, in progress, targeting Nature Computational Science] *Neptune: An open-source platform for microfluidic design automation with a new programming language and in-context LLM assistance.*